

**OCEANSENSE: AN IOT-BASED HYBRID SYSTEM FOR
REAL-TIME DETECTION AND TRACKING OF
MARINE DEBRIS**

25-26J-002

Project Proposal Report

Jayasak D L C – IT22908360

Rathnayake G R M S D – IT22228130

Abeysinghe P T G S R – IT22253576

Dissanayaka D M T N – IT22169730

Supervisor – Ms. Dinithi Pandithage

Co-Supervisor – Ms. Narmada Gamage

BSc (Hons) in Information Technology Specializing in Computer
Systems & Network Engineering

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

**OCEANSENSE: AN IOT-BASED HYBRID SYSTEM FOR
REAL-TIME DETECTION AND TRACKING OF
MARINE DEBRIS**

25-26J-002

Project Proposal Report

BSc (Hons) in Information Technology Specializing in Computer
Systems & Network Engineering

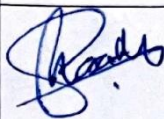
Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

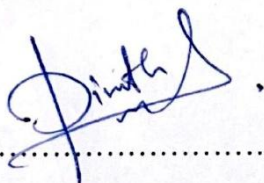
August 2025

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

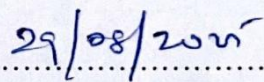
Name	Student ID	Signature
Jayasak D L C	IT22908360	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

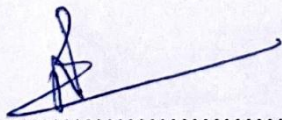


Signature of the Supervisor

(Ms. Dinithi Pandithage)

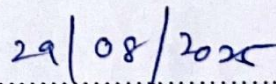


Date



Signature of the Co-Supervisor

(Ms. Narmada Gamage)



Date

ABSTRACT

Marine debris poses a serious threat to ocean ecosystems, navigation safety, and coastal environments. To address this challenge, this project proposes a decentralized swarm-based marine debris detection system using low-cost IoT technologies. The system consists of multiple floating autonomous nodes that collaboratively monitor sea surface debris using onboard sensors and vision-based detection models.

LoRa is employed as the **primary node-to-node communication method**, enabling long-range, low-power data exchange within the swarm network. To enhance system reliability in dynamic marine environments, a **location-aware self-healing mechanism** is introduced. Unlike conventional fault tolerance approaches that focus solely on communication redundancy, the proposed self-healing mechanism enables a disconnected node to autonomously recover its position and rejoin the swarm.

When a node loses LoRa connectivity, it retrieves its last known location using GSM-assisted communication and navigates back toward the swarm using GPS, compass-based heading estimation, and onboard propulsion. This ensures continuous area coverage, prevents data loss, and minimizes human intervention. A prototype implementation using Arduino-based nodes demonstrates the feasibility of the proposed approach. The system is scalable and suitable for long-term marine monitoring applications.

TABLE OF CONTENTS

DECLARATION	I
ABSTRACT	II
TABLE OF CONTENTS	III
1 INTRODUCTION.....	1
1.1 RESEARCH PROBLEM.....	3
2 OBJECTIVES	4
2.1 MAIN OBJECTIVE:	4
2.2 SPECIFIC OBJECTIVES:	4
3 METHODOLOGY	5
4 PROJECT REQUIREMENTS.....	6
4.1 FUNCTIONAL REQUIREMENTS:	6
4.2 NON-FUNCTIONAL REQUIREMENTS:	7
5 BUDGET AND JUSTIFICATION	8
REFERENCES.....	9
APPENDICES.....	10
5.1 GANTT CHART	10

1 INTRODUCTION

Marine debris has become a major environmental issue, causing severe harm to marine ecosystems, wildlife, and coastal economies. Plastics and other floating waste materials accumulate over large ocean areas, making manual monitoring and cleanup operations inefficient and costly. As a result, autonomous marine monitoring systems have gained significant attention as a sustainable solution for large-scale ocean observation and debris detection.

Swarm-based autonomous systems provide an effective approach for marine monitoring by enabling multiple low-cost floating nodes to collaboratively observe large areas. In such systems, each node operates independently while coordinating with neighboring nodes to ensure efficient area coverage. However, reliable communication among swarm nodes remains a critical challenge in marine environments due to long distances, environmental interference, and continuous node drift caused by ocean currents.

In this project, LoRa is employed as the primary communication method for node-to-node interaction within the marine swarm network. LoRa technology offers long-range, low-power wireless communication, making it highly suitable for marine surface applications where energy efficiency and communication range are essential. Through LoRa-based communication, swarm nodes exchange status and location information to maintain coordinated operation.

Despite the advantages of LoRa, communication failures may still occur due to environmental conditions or physical separation of nodes. Such failures can cause individual nodes to become disconnected from the swarm, leading to coverage gaps and reduced monitoring efficiency. To address this issue, the proposed system introduces a self-healing mechanism that allows a disconnected node to autonomously recover and rejoin the swarm.

Self-healing in this system does not involve changing the primary communication method. Instead, when a node loses LoRa connectivity, GSM is used only as a supporting mechanism to retrieve the node's last known location and time information. Using this information along with onboard GPS positioning, digital compass-based heading estimation, and motor-driven propulsion, the node autonomously navigates back toward the swarm region. Once LoRa communication is restored, the node resumes normal swarm operation.

By integrating LoRa-based swarm communication with a location-aware self-healing mechanism, the proposed system enhances reliability, fault tolerance, and continuous area coverage in dynamic marine environments. The system is designed using low-cost IoT hardware and demonstrated through a prototype implementation, making it a scalable and practical solution for marine debris monitoring applications.

Research Gap

Existing marine debris monitoring systems primarily rely on manual surveys, satellite imagery, or single autonomous platforms, which are limited in scalability, cost efficiency, and real-time responsiveness. Recent research has explored swarm-based autonomous systems for large-area marine monitoring; however, several critical gaps remain unresolved.

Firstly, many existing swarm-based marine IoT systems depend on centralized control architectures, where the failure or disconnection of a single node can lead to reduced coverage and loss of system reliability. These systems often assume continuous communication availability, which is unrealistic in dynamic marine environments affected by node drift, environmental interference, and long-distance separation.

Secondly, while LoRa has been widely studied for long-range marine communication, most existing works utilize LoRa solely for uplink data transmission to gateways, rather than for decentralized node-to-node swarm coordination. There is limited research on using LoRa as the primary communication backbone within autonomous marine swarms.

Furthermore, existing fault-tolerance approaches in marine IoT systems mainly focus on communication redundancy or alert-based recovery, where a disconnected node simply reports failure or waits for manual intervention. Very few systems address physical recovery of the node's position, especially in floating marine platforms where node drift can significantly affect coverage.

Additionally, self-healing mechanisms in current swarm systems are often limited to network-level recovery and do not incorporate location-aware autonomous navigation using low-cost hardware. The integration of last-known location data, GPS-based positioning, compass-guided navigation, and onboard propulsion for autonomous recovery remains underexplored in marine swarm applications.

Therefore, there exists a research gap in the development of a decentralized, location-aware self-healing mechanism for LoRa-based marine swarm systems, where disconnected nodes can autonomously navigate back to the swarm to preserve area coverage and system reliability without relying on continuous communication or human intervention.

1.1 Research Problem

Marine swarm-based monitoring systems are highly susceptible to communication failures caused by environmental interference, node drift, and dynamic sea conditions. In such scenarios, the loss of communication between swarm nodes can result in incomplete area coverage, loss of monitoring data, and reduced system reliability. Conventional systems rely heavily on centralized control or continuous communication availability, which is impractical in remote marine environments.

Therefore, a decentralized and autonomous recovery mechanism is required to ensure that swarm nodes can recover from communication failures and rejoin the network without external intervention.

2 OBJECTIVES

2.1 Main Objective:

To design and implement a decentralized marine debris detection swarm system with autonomous self-healing capabilities using low-cost IoT hardware.

2.2 Specific Objectives:

To implement **LoRa as the primary node-to-node communication method** for swarm coordination.

To design a **location-aware self-healing mechanism** that enables disconnected nodes to autonomously rejoin the swarm.

To utilize **GSM communication only as a fallback mechanism** for retrieving last-known location information.

To integrate **GPS and compass sensors** for autonomous navigation and heading correction.

To control propellers using a motor driver to enable physical movement of nodes on the sea surface.

To demonstrate the feasibility of the system using a **two-node prototype implementation**.

To ensure continuous area coverage and fault tolerance in dynamic marine environments.

3 METHODOLOGY

Each swarm node consists of an Arduino-based controller, a LoRa transceiver for primary communication, a GSM module for fallback assistance, a GPS module for location tracking, a digital compass for heading estimation, and motor-driven propellers for navigation. During normal operation, nodes communicate continuously using LoRa to exchange status and location information.

If a node fails to receive LoRa messages within a predefined time interval, it identifies itself as disconnected from the swarm. The node then initiates the self-healing process, during which GSM communication is temporarily activated to retrieve the last known location and timestamp. Using current GPS data and compass-based direction estimation, the node autonomously navigates back toward the swarm region until LoRa communication is restored.

Self-healing in the proposed system refers to the autonomous recovery of a disconnected node's position, rather than a change in communication protocol. When a node loses LoRa connectivity, it does not immediately rely on GSM for continuous communication. Instead, GSM is used selectively to retrieve the node's last known location information.

The node then calculates the direction toward the swarm using GPS and compass data and physically moves toward the target location using onboard propellers. Once LoRa communication is re-established, the self-healing process is terminated and the node resumes normal swarm operation. This approach ensures resilience and fault tolerance while minimizing power consumption and communication overhead.

The scope of this project focuses on the design and implementation of a prototype marine swarm node with location-aware self-healing capability. The system includes LoRa-based primary communication, GSM-assisted fallback for location retrieval, GPS-based positioning, compass-based navigation, and motor-driven propulsion.

Due to hardware and time constraints, the implementation is limited to a two-node demonstration; however, the proposed architecture is scalable to larger swarm networks. Advanced collision avoidance, high-precision navigation, and large-scale field deployment are beyond the scope of this project.

4 PROJECT REQUIREMENTS

4.1 Functional Requirements:

FR1. Each swarm node shall capture sensor data related to marine debris detection and process it locally using onboard computation modules. (Image acquisition and detection accuracy are handled by a separate subsystem and are outside the scope of this work.)

FR2. Each node shall exchange status, location, and detection metadata with neighboring nodes using LoRa as the primary node-to-node communication method.

FR3. Each node shall continuously monitor LoRa connectivity and detect communication loss within a predefined timeout interval.

FR4. Upon detection of LoRa communication failure, the node shall initiate a self-healing mechanism to autonomously recover its position and rejoin the swarm network.

FR5. During self-healing, the node shall use GSM communication only to retrieve its last known location and time information, and not for continuous swarm communication.

FR6. The node shall determine its current position using GPS and compute the direction toward the swarm using compass-based heading estimation.

FR7. The node shall control onboard propellers through a motor driver to physically navigate toward the last known swarm location.

FR8. Once LoRa communication is restored, the node shall terminate the self-healing process and resume normal swarm operation.

4.2 Non-Functional Requirements:

NFR1 (Energy Efficiency):

The system shall prioritize low-power operation by using LoRa as the primary communication method and activating GSM only during self-healing scenarios.

NFR2 (Reliability):

The swarm network shall tolerate temporary communication loss or node drift by enabling autonomous recovery of disconnected nodes without human intervention.

NFR3 (Scalability):

The proposed architecture shall be scalable to multiple swarm nodes without requiring centralized control or manual reconfiguration. The two-node prototype demonstrates the feasibility of the scalable design.

NFR4 (Robustness):

The system shall maintain continuous area coverage by enabling nodes to autonomously return to the swarm after disconnection caused by environmental factors.

NFR5 (Latency):

Communication and recovery mechanisms shall ensure timely rejoining of disconnected nodes to minimize data loss and coverage gaps in marine monitoring operations.

NFR5 (Latency): Critical detection alerts (e.g. large debris cluster) must reach the central dashboard within 5 minutes. Routine data may have higher delays.

5 BUDGET AND JUSTIFICATION

Item	Quantity	Unit Cost(Rs)	Total Cost(Rs)
ESP32 Development Boards	3	1300	3900
Wave-Energy Harvester	2	2500	5000
Solar Panels + Batteries	4	1000	4000
LoRa Module	3	4500	13500
Camera Modules (low-light)	4	1200	4800
Raspberry Pi 4	3	18000	54000
Miscellaneous (cables, buoys)	N/A	N/A	12000
Total	N/A	N/A	97200

REFERENCES

- [1] M. Kunwar, "IoT in Education: From Smart Classrooms to Connected Campuses," 26 November 2024. [Online]. Available: <https://www.hashstudioz.com/blog/iot-in-education-from-smart-classrooms-to-connected-campuses/>. [Accessed 11 August 2025].
- [2] G. Sarraf, "Introduction to TinyML on Microcontrollers: Bringing AI to the Edge," 19 May 2025. [Online]. Available: <https://thinkrobotics.com/blogs/learn/introduction-to-tinyml-on-microcontrollers-bringing-ai-to-the-edge>. [Accessed 10 August 2025].
- [3] B. Hugar, I. M, K. S. Sree, M. Deepa and S. Devale, "Moodscape: Adaptive Environment Based on User," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, vol. 13, no. 5, p. 7, 2025.
- [4] Y. Cho, H. Lee, A.-S. Choi and M. Sung, "Development and Application of IoT-Based Emotional Lighting System: Focusing on Emotion Analysis Using Instant Message Data," *LEUKOS - The Journal of the Illuminating Engineering Society*, vol. 21, no. 2, pp. 182-195, 2024.
- [5] L. J, L. K and Z. H, "Global trends in marine plastic pollution and its ecological impacts," *IEEE Access*, vol. 10, p. 55203–55216, 2022.
- [6] K. L. H. Z. J. Li, "Global trends in marine plastic pollution and its ecological impacts," *IEEE Access*, vol. 10, p. 55203–55216, 2022.
- [7] L. L. R. Diamant, "Underwater acoustic communications: Challenges and future trends," *IEEE Journal of Oceanic Engineering*, vol. 42, no. 4, p. 1447–1464, 2017.

APPENDICES

Appendix A:

5.1 Gantt Chart

Gantt Chart

Activity	1	2	3	4	5	6	7	8	9	10
Background Research										
System Design										
Implementation										
Testing & Evaluation										
Report Writing										

Appendix B: System Diagram

